

JAMESTOWN 38 WSW, ND, USAWMO: **727510**Lat: **46.7702N**Lon: **99.4778W**Elev: **1920**StdP: **13.70**Time Zone: **-6.00 (NAC)**Period: **08-19**WBAN: **54937****Annual Heating, Humidification, and Ventilation Design Conditions**

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-19.1	-15.1	-24.1	1.6	-18.5	-19.8	2.0	-14.4	31.1	15.7	28.6	15.0	5.6	N/A	0.616

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	21.2	87.7	70.3	84.3	68.3	81.4	66.9	73.3	82.9	71.1	81.0	69.2	78.9	9.4	N/A

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
70.2	119.5	78.7	68.0	110.3	76.9	65.8	102.3	74.5	38.3	82.5	36.1	80.8	34.5	78.9	83.3

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 25.9	22.2	19.4	-27.1	95.3	4.5	4.0	-30.3	98.2	-32.9	100.5	-35.5	102.7	-38.7	105.6		
(5)			-26.4	77.5	4.2	2.8	-29.4	79.6	-31.9	81.2	-34.3	82.8	-37.3	84.9		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	40.3	9.9	11.0	25.7	39.4	53.1	63.2	68.8	66.8	58.4	43.1	28.8	13.5
	DBStd	23.66	14.16	13.09	14.20	10.80	8.99	6.42	5.66	5.98	8.12	9.53	11.95	13.20
	HDD50	5554	1243	1092	759	340	72	3	0	0	19	256	638	1132
	HDD65	9356	1708	1512	1218	768	379	105	26	49	228	680	1086	1597
	CDD50	2011	0	0	6	21	169	397	583	520	270	43	2	0
	CDD65	340	0	0	0	0	11	50	144	104	29	2	0	0
	CDH74	3336	0	0	1	9	145	448	1336	1031	340	26	0	0
	CDH80	945	0	0	0	1	23	86	399	342	88	6	0	0
	WSAvg	8.2	8.7	8.4	9.0	10.0	9.3	7.6	6.3	6.3	7.4	9.0	8.2	8.1
	PrecAvg	20.1	0.5	0.4	0.7	1.1	2.9	3.5	3.6	2.1	2.1	1.7	0.6	0.5
(15) Precipitation	PrecMax	27.6	1.2	1.3	2.7	2.2	6.4	7.2	10.1	5.0	4.0	4.3	2.2	1.2
	PrecMin	15.2	0.0	0.0	0.1	0.3	0.3	1.8	0.7	0.4	0.2	0.1	0.0	0.1
	PrecStd	3.3	0.4	0.4	0.7	0.6	2.0	1.3	2.1	1.3	1.1	1.2	0.6	0.3
	PrecStd	3.3	0.4	0.4	0.7	0.6	2.0	1.3	2.1	1.3	1.1	1.2	0.6	0.3
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4% DB	42.3	43.7	67.9	74.1	83.6	86.1	91.6	93.5	88.8	77.2	63.4	45.2	
	0.4% MCWB	35.7	37.1	50.9	54.4	62.4	69.8	72.7	69.3	68.5	59.9	50.4	38.6	
	2% DB	36.9	38.1	60.5	66.8	78.0	82.5	87.8	87.7	82.0	69.2	55.4	38.3	
	2% MCWB	33.0	34.2	49.7	51.1	60.4	66.9	71.5	69.0	64.5	54.6	46.2	34.1	
	5% DB	33.6	34.2	52.1	61.9	73.9	79.0	84.4	83.6	77.7	63.5	49.3	34.4	
	5% MCWB	31.2	31.6	43.4	47.9	58.8	65.0	69.9	68.2	62.4	51.7	41.7	31.4	
	10% DB	30.4	30.4	43.7	57.1	69.4	76.0	81.4	79.7	73.3	58.3	44.2	30.7	
	10% MCWB	28.6	28.5	38.7	45.2	56.4	63.6	68.6	65.8	60.9	49.4	38.9	28.9	
	0.4% WB	37.2	38.9	56.6	56.0	66.2	73.2	78.6	74.3	71.0	61.1	51.0	39.3	
	0.4% MCDB	44.3	43.8	62.4	69.7	77.0	82.5	85.5	85.2	82.7	74.3	63.1	44.6	
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	2% WB	33.7	35.0	49.8	51.7	64.1	70.0	74.7	71.5	68.1	57.2	45.8	35.1	
	2% MCDB	36.8	38.1	62.6	64.6	75.4	78.7	82.9	82.6	77.6	67.6	54.5	38.9	
	5% WB	31.9	32.5	44.2	48.6	61.1	67.7	72.3	69.8	65.3	53.3	41.5	32.2	
	5% MCDB	34.4	35.0	53.3	60.2	69.3	77.1	81.4	79.7	74.6	62.5	46.8	35.0	
	10% WB	29.3	29.2	39.5	45.9	58.5	65.5	70.1	68.0	62.5	50.1	38.5	29.6	
	10% MCDB	31.4	31.4	45.5	55.3	66.7	74.3	80.1	77.3	70.9	58.2	43.2	31.7	
(35) Mean Daily Temperature Range	MDBR	16.9	16.9	17.3	19.8	21.5	20.2	21.2	22.1	22.0	18.9	17.8	15.1	
	5% DB	19.6	19.3	28.2	29.4	27.0	24.5	24.7	26.7	27.9	28.0	25.0	18.6	
	5% WB	17.0	15.6	17.1	16.1	13.4	13.0	11.5	11.9	13.5	15.3	15.7	14.3	
	MCDBR	19.3	19.4	27.9	28.7	21.9	21.9	21.3	23.4	24.6	25.8	23.5	17.5	
	MCWBR	16.6	15.5	17.4	16.0	11.4	12.7	12.5	11.5	13.1	15.1	15.3	13.9	
	MCWBR	16.6	15.5	17.4	16.0	11.4	12.7	12.5	11.5	13.1	15.1	15.3	13.9	
(40) Clear-Sky Solar Irradiance	taub	0.296	0.300	0.298	0.358	0.370	0.368	0.374	0.375	0.345	0.296	0.267	0.252	
	taud	2.256	2.302	2.441	2.296	2.340	2.400	2.397	2.376	2.433	2.551	2.509	2.455	
	Ebn at Noon	248	275	296	284	284	284	281	276	275	278	266	261	
	Edh at Noon	25	30	30	39	38	36	36	35	30	23	19	18	
(44) All-Sky Solar Radiation	RadAvg	536	892	1248	1573	1797	1998	2073	1739	1319	829	525	398	
	RadStd	47	91	99	124	136	134	100	137	86	104	67	35	

Historical Trends

	DBAvg	Heating		
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